

*Achievements of School Educational Data
Analysis With SQL & Power BI*

By Hassan Nezamoleslam

In this study, multiple visualisations were developed to support data-driven analysis of student performance.

These outputs can be used in meetings with parents, teachers, and academic advisors to evaluate students quantitatively, rather than relying solely on qualitative assessments.

1. Analysis of average grades across all students for each subject in both mock exams and final exams.

This enables tracking of performance trends (improvement or decline) and evaluation of the reliability of mock exams.

2. A more detailed analysis can be conducted at the individual student level, allowing performance tracking for each student separately.

3. A comprehensive overview of each student's performance across all subjects can be monitored, providing a deeper level of insight compared to the previous stage.

4. Identification of top-performing and underperforming students in each subject, enabling tailored academic strategies and learning pathways.

5. Simultaneous monitoring of an individual student's performance using multiple visualisations, along with comparison against peers.

6. Evaluation of each student's performance relative to the class average, highlighting whether they are above or below the overall level.

7. Ranking of average grades across subjects and student groups (e.g., by class or gender), enabling better understanding of performance patterns and fostering healthy academic competition.

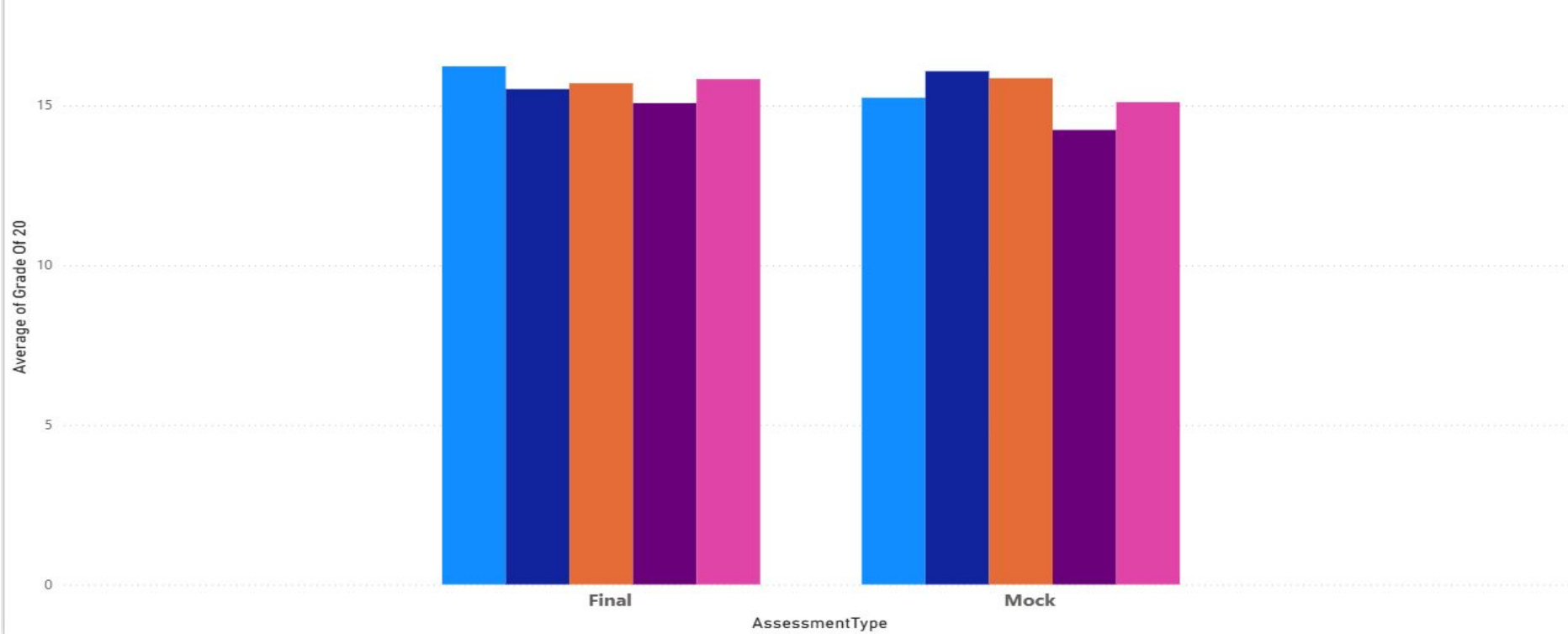
8. These visualisations and findings represent a simplified sample. More advanced scenarios and analyses can be developed to extend the scope and depth of insights.

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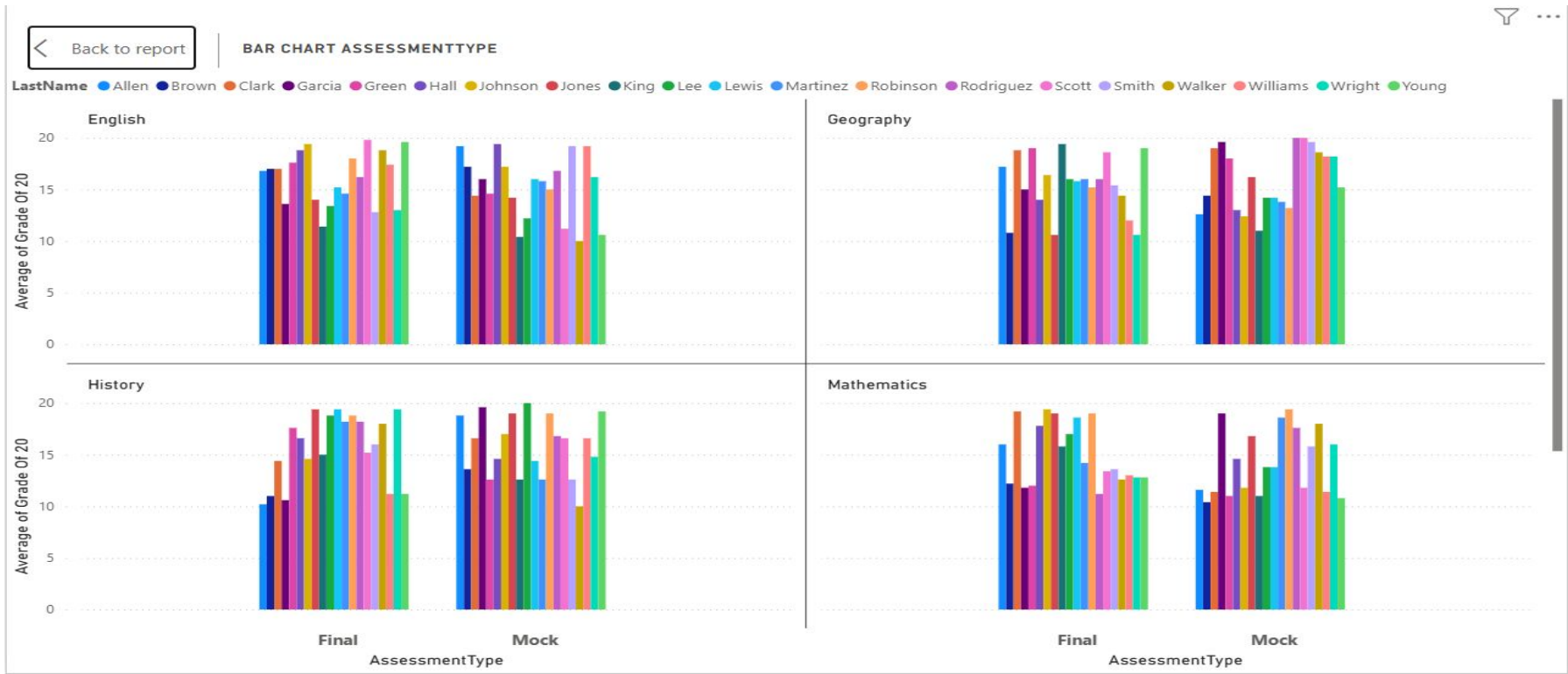
GARDES (MOCK +FINAL) BY SUBJECTS



SubjectName ● English ● Geography ● History ● Mathematics ● Science



Average Grades & AssessmentType(Mock,Final), Subject Legend



Grades & SubjectName For Every Student

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MATRIX OF EXAM

AssessmentType	Allen	Brown	Clark	Garcia	Green	Hall	Johnson	Jones	King	Lee	Lewis	Martinez	Robinson	Rodriguez	Scott	Smith	Walker	Williams	Wright	Young	Total
Final	14.48	13.00	17.60	13.08	15.96	16.72	17.36	15.96	14.32	15.32	17.72	15.88	18.00	15.80	16.00	15.08	16.36	13.88	14.84	15.88	15.66
English	16.80	17.00	17.00	13.60	17.60	18.80	19.40	14.00	11.40	13.40	15.20	14.60	18.00	16.20	19.80	12.80	18.80	17.40	13.00	19.60	16.22
Geography	17.20	10.80	18.80	15.00	19.00	14.00	16.40	10.60	19.40	16.00	15.80	16.00	15.20	16.00	18.60	15.40	14.40	12.00	10.60	19.00	15.51
History	10.20	11.00	14.40	10.60	17.60	16.60	14.60	19.40	15.00	18.80	19.40	18.20	18.80	18.20	15.20	16.00	18.00	11.20	19.40	11.20	15.69
Mathematics	16.00	12.20	19.20	11.80	12.00	17.80	19.40	19.00	15.80	17.00	18.60	14.20	19.00	11.20	13.40	13.60	12.60	13.00	12.80	12.80	15.07
Science	12.20	14.00	18.60	14.40	13.60	16.40	17.00	16.80	10.00	11.40	19.60	16.40	19.00	17.40	13.00	17.60	18.00	15.80	18.40	16.80	15.82
Mock	14.60	14.20	15.08	18.68	13.48	15.04	14.20	16.48	12.32	15.28	15.44	15.52	15.80	18.20	15.48	16.12	15.08	16.04	15.60	13.32	15.30
English	19.20	17.20	14.40	16.00	14.60	19.40	17.20	14.20	10.40	12.20	16.00	15.80	15.00	16.80	11.20	19.20	10.00	19.20	16.20	10.60	15.24
Geography	12.60	14.40	19.00	19.60	18.00	13.00	12.40	16.20	11.00	14.20	14.20	13.80	13.20	20.00	20.00	19.60	18.60	18.20	18.20	15.20	16.07
History	18.80	13.60	16.60	19.60	12.60	14.60	17.00	19.00	12.60	20.00	14.40	12.60	19.00	16.80	16.60	12.60	10.00	16.60	14.80	19.20	15.85
Mathematics	11.60	10.40	11.40	19.00	11.00	14.60	11.80	16.80	11.00	13.80	13.80	18.60	19.40	17.60	11.80	15.80	18.00	11.40	16.00	10.80	14.23
Science	10.80	15.40	14.00	19.20	11.20	13.60	12.60	16.20	16.60	16.20	18.80	16.80	12.40	19.80	17.80	13.40	18.80	14.80	12.80	10.80	15.10
Total	14.54	13.60	16.34	15.88	14.72	15.88	15.78	16.22	13.32	15.30	16.58	15.70	16.90	17.00	15.74	15.60	15.72	14.96	15.22	14.60	15.48

Fast Look Of Student's Situation

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THE FIVE TOP STUDENTS

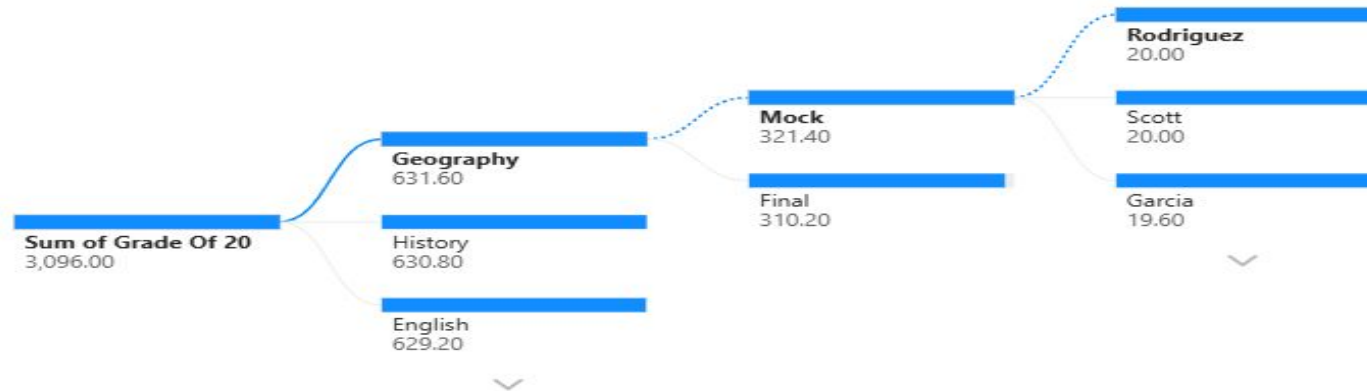
SubjectName x

Geography

AssessmentType x

Mock

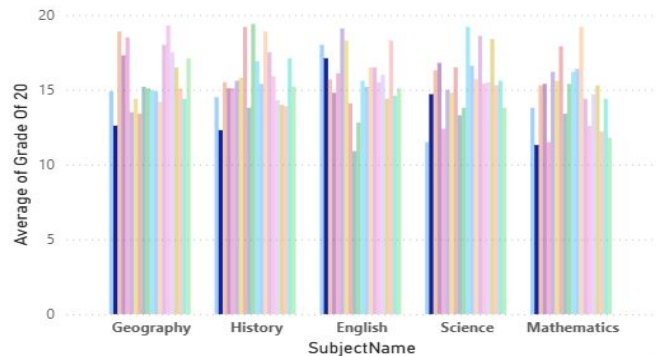
LastName x



Rank Of Top Students Base on Subject And AssessmentType

Student's Grades

LastName ● Allen ● Brown ● Clark ● Garcia ● Green ● Hall ● Johnson



Ask Me

⚠ Q&A will be retiring in December 2026.

Remove Q&A

Grace (JoinAcademicGradesSubjectStudent (2) first name)

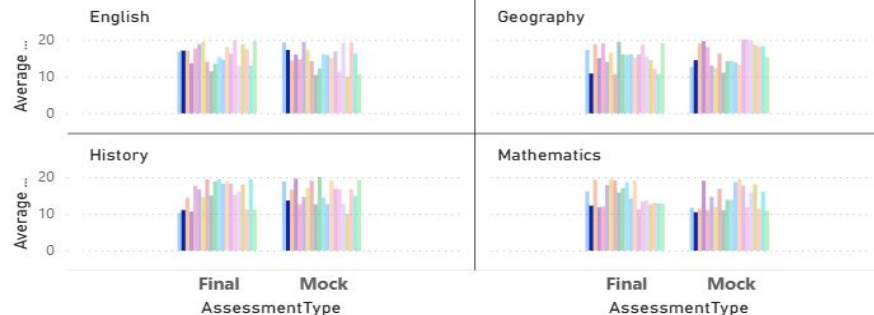
StudentID AssessmentDate AssessmentType FirstName LastName Score

Matrix Of Exam

AssessmentType	Allen	Brown	Clark	Garcia	Green	Hall	Johnson	Jones	King	Lee	Lewis
Final	14.48	13.00	17.60	13.08	15.96	16.72	17.36	15.96	14.32	15.32	17.72
English	16.80	17.00	17.00	13.60	17.60	18.80	19.40	14.00	11.40	13.40	15.20
Geography	17.20	10.80	18.80	15.00	19.00	14.00	16.40	10.60	19.40	16.00	15.80
History	10.20	11.00	14.40	10.60	17.60	16.60	14.60	19.40	15.00	18.80	19.40
Mathematics	16.00	12.20	19.20	11.80	12.00	17.80	19.40	19.00	15.80	17.00	18.60
Science	12.20	14.00	18.60	14.40	13.60	16.40	17.00	16.80	10.00	11.40	19.60
Mock	14.60	14.20	15.08	18.68	13.48	15.04	14.20	16.48	12.32	15.28	15.44
English	19.20	17.20	14.40	16.00	14.60	19.40	17.20	14.20	10.40	12.20	16.00
Geography	12.60	14.40	19.00	19.60	18.00	13.00	12.40	16.20	11.00	14.20	14.20
History	18.80	13.60	16.60	19.60	12.60	14.60	17.00	19.00	12.60	20.00	14.40
Mathematics	11.60	10.40	11.40	19.00	11.00	14.60	11.80	16.80	11.00	13.80	13.80
Science	10.80	15.40	14.00	19.20	11.20	13.60	12.60	16.20	16.60	16.20	18.80
Total	14.54	13.60	16.34	15.88	14.72	15.88	15.78	16.22	13.32	15.30	16.58

Bar Chart AssessmentType

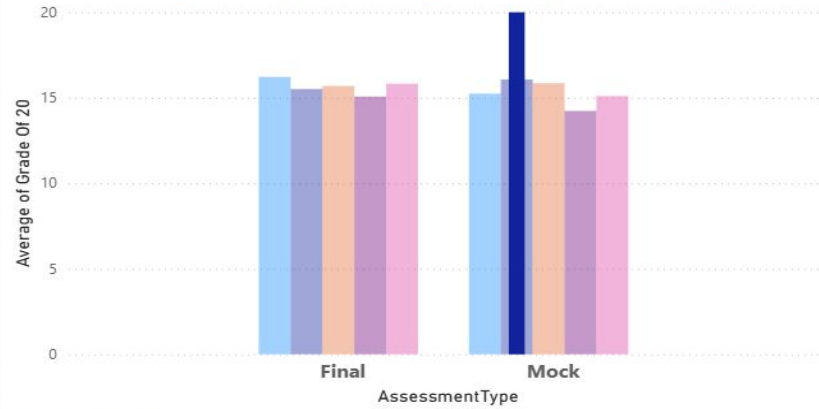
LastName ● Allen ● Brown ● Clark ● Garcia ● Green ● Hall ● Johnson ● Jones ● King ● Lee ● Lewis



Trace Student's Situation In the different charts Simultaneously

Gardes (Mock +Final) By Subjects

SubjectName ● English ● Geography ● History ● Mathematics ● Science



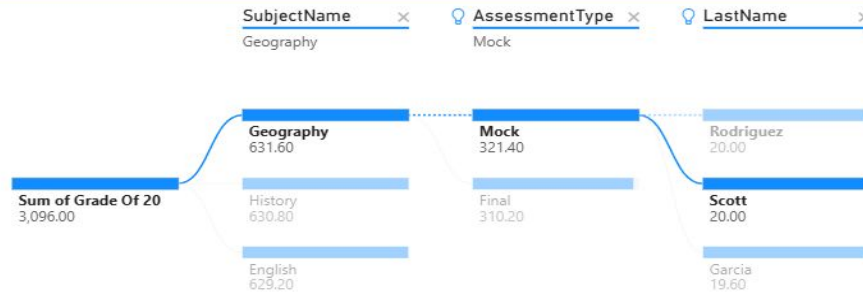
Quick Look Of The Exams

AssessmentType	Geography	Total
Mock	20.00	20.00
Total	20.00	20.00

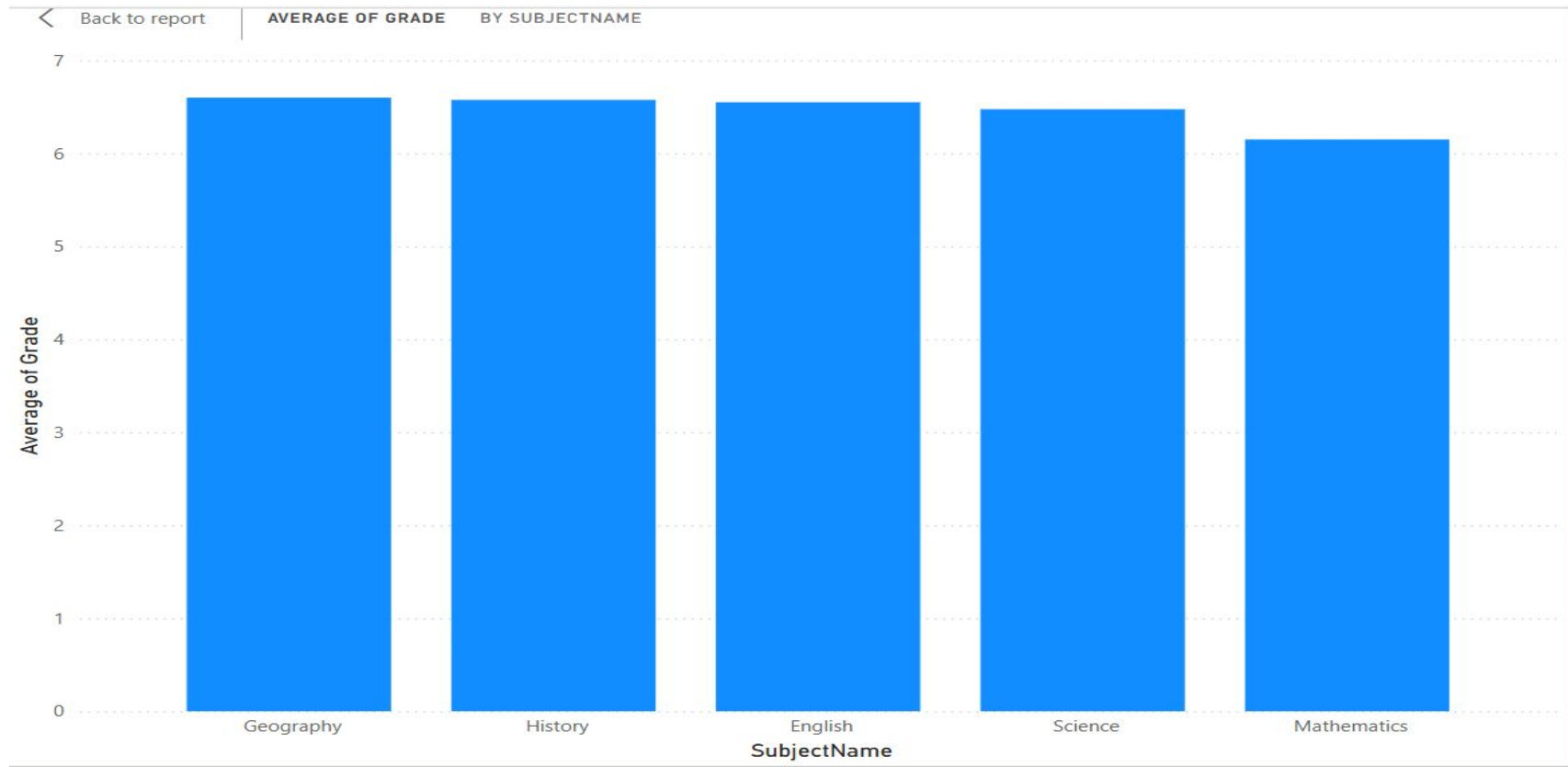
Analysis Of AssessmentType



The Five Top Students



Compare A Student's Grades with The Average



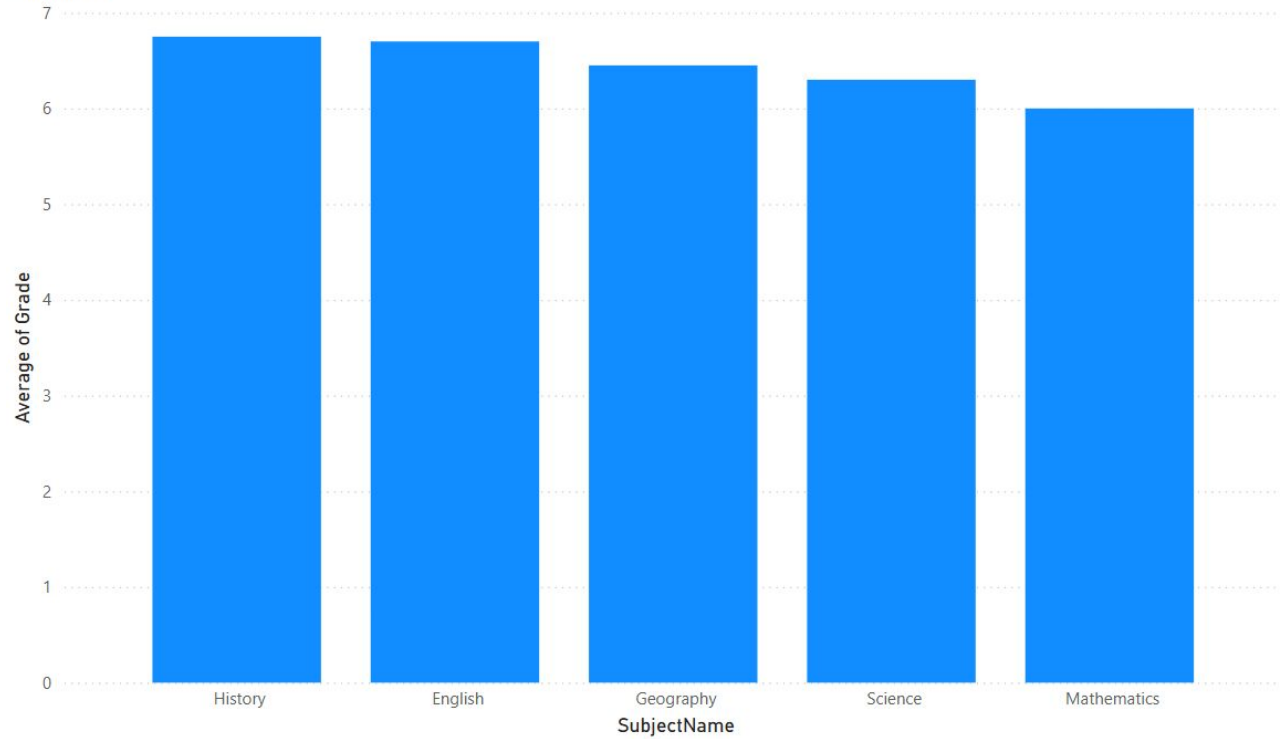
Sort The Average Grades For Different SubjectName

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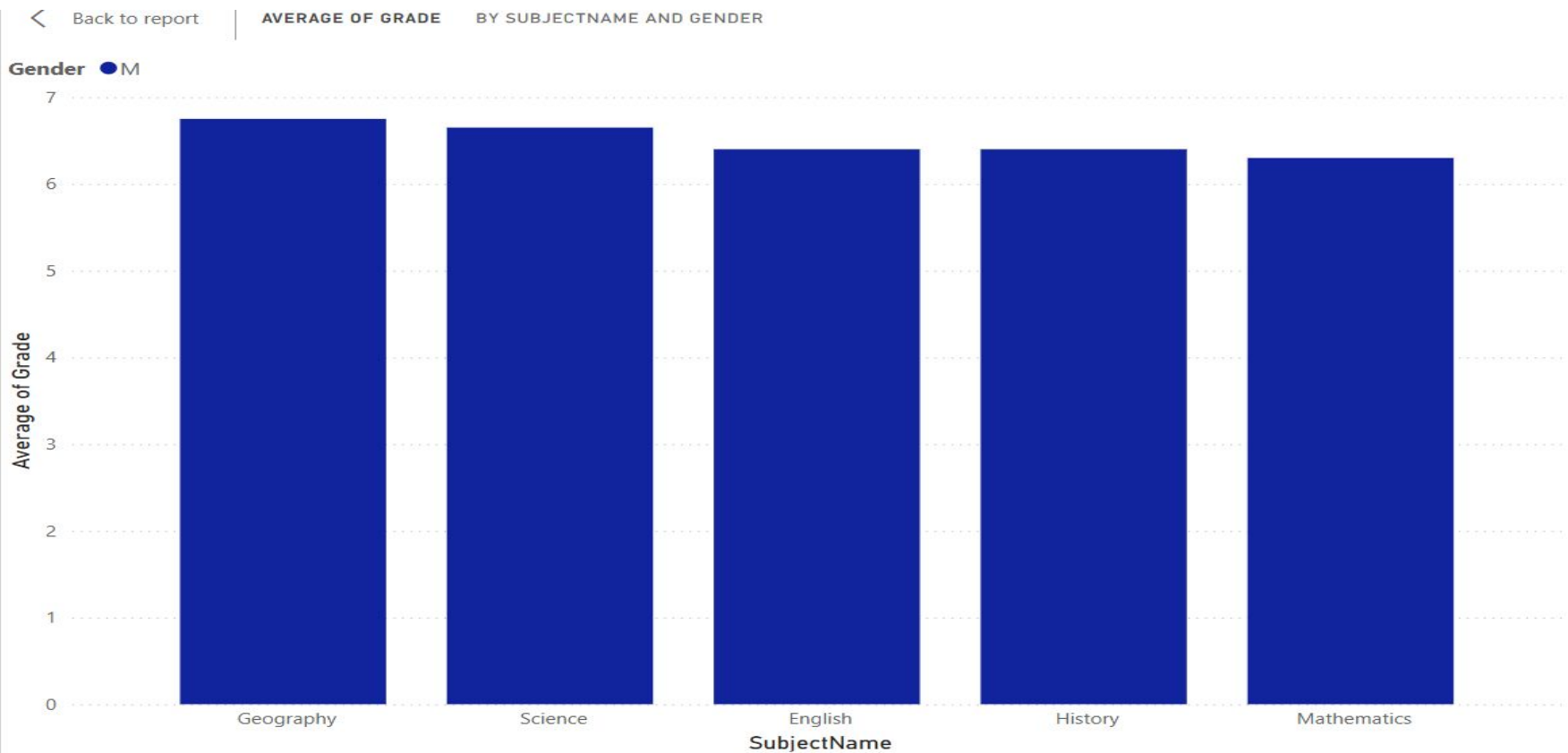
AVERAGE OF GRADE

BY SUBJECTNAME AND GENDER

Gender ● F



Sort The Average Grades Base On SubjectName&Gender 'Girls'



Sort The Average Grades Base On SubjectName&Gender 'Boys'